

Documentation for Noah-MP

SMAP DA Evaluation

Framework

This directory (`/scripts/postproc/postproc_01`) contains the analytical scripts to generate a set of 14 scientific figures evaluating the impact of Soil Moisture Active Passive (SMAP) Data Assimilation (DA) on the Noah-MP land surface model over the Sebou basin.

Below is the detailed description of each figure, including its scientific objective, the required datasets, their sources, and their **current availability** in the workspace.

Figure 1: Study Domain and Land Cover

- **Objective / Message:** Contextualize the physiographic and hydroclimatic characteristics of the Sebou basin.

- **Data Used:**
 - **Relief:** SRTM DEM 30m (Source: NASA, Static) - *Assumed Available (LIS Input)*
 - **Land Cover:** MODIS IGBP (Source: NASA, Static) - *Assumed Available (LIS Input)*
 - **Basin & River Network:** HydroSHEDS / MERIT (Source: WWF / Yamazaki et al., Static) - *Assumed Available*
 - **In Situ Stations:** GPS Coordinates (Source: Local Agencies, Static) - *Assumed Available*
- **Status:** Ready.

Figure 2: Conceptual Modeling Framework

- **Objective / Message:** Clearly describe the experimental protocol (Forcings -> LIS-Noah-MP -> OL vs DA -> HyMAP).
- **Data Used:** None (Schematic generation using Python `graphviz` or `matplotlib` patches).
- **Status:** Ready.

Figure 3: Impact of Assimilation on Soil Moisture

- **Objective / Message:** Identify corrections brought by DA according to seasons (winter vs summer).

- **Data Used:**
 - **Noah-MP Output:** Soil moisture layer 1 (0-5cm) from OL and DA. (Temporal Freq: Daily/3-Hourly, Source: Local LIS Simulations).
- **Status:** Partially Ready. The script is functional using the `3days` test dataset (`experiments/3days/`), but full seasonal analysis requires the multi-year spin-up/assimilation runs to be completed.

Figure 4: Impact on Total Runoff and Link with Irrigation

- **Objective / Message:** Demonstrate if assimilation indirectly captures the effect of irrigation and modifies the runoff.
- **Data Used:**
 - **Noah-MP Output:** Total Runoff ($Q_s + Q_{sb}$) from OL and DA. (Temporal Freq: Daily/3-Hourly, Source: Local Simulations).
 - **Irrigation Map:** Global Map of Irrigation Areas (GMIA) v5. (Temporal Freq: Static, Source: FAO).
- **Status:** Ready. GMIA data found at `/data/land_params/GMIA/gmia_v5_aei_pct.asc` .

Figure 5: Runoff Decomposition

- **Objective / Message:** Determine which hydrological component (surface runoff vs baseflow) is most sensitive to assimilation.
- **Data Used:**
 - **Noah-MP Output:** Surface runoff (Q_s), Baseflow (Q_{sb}), Soil Moisture from OL and DA.
 - **Forcing:** IMERG Precipitation.
- **Status:** Partially Ready (Requires full simulation for meaningful correlation).

Figures 6 to 9: Multi-Model Intercomparison (GLDAS)

- **Objective / Message:** Position the local LIS simulation results relative to global reference models (GLDAS NOAH, VIC, CLSM).
- **Data Used:**
 - **Noah-MP Output:** Soil Moisture, Runoff, Evapotranspiration.
 - **GLDAS Products:** GLDAS_NOAH025_3H v2.1, GLDAS_VIC10_3H v2.1, GLDAS_CLSM10_3H v2.1, GLDAS_CLSM025_DA1_D v2.2. (Temporal Freq: 3-Hourly/Daily, Source: GES DISC).

- **Status: MISSING.** GLDAS files are not yet downloaded in the `/data` directory. The scripts `( fig06_09_gldas_intercomparison.py )` are provided as skeletons that will process the files once they are downloaded.

Figures 10 to 12: Validation of Routed Streamflow

- **Objective / Message:** Quantify the improvement brought by DA on streamflow dynamics (hydrographs, KGE, NSE).
- **Data Used:**
 - **HyMAP Output:** Routed discharge from OL and DA. (Temporal Freq: Daily, Source: Local HyMAP offline runs).
 - **In Situ Observations:** Gauging station discharge. (Temporal Freq: Daily, Source: Local Agencies).
- **Status: MISSING.** Both HyMAP routing outputs and in-situ streamflow observations are not yet available for the `3days` test. The scripts are provided as skeletons.

Figures 13 and 14: Spatial Analysis of the Impact of Assimilation

- **Objective / Message:** Identify factors controlling the hydrological response to DA (basin size, precipitation, irrigation).
- **Data Used:**
 - **HyMAP / Noah-MP Output:** Relative streamflow/runoff changes.
 - **IMERG Precipitation:** Total precipitation.
 - **GMIA Irrigation:** Irrigation percentage.
- **Status:** MISSING. Depends on routed discharge (HyMAP), which is not yet available. Scripts provided as skeletons.

Summary Table

Figure	message	data used from simulation (variables) if any	data used from observation (if any)
Figure 1	Contextualize the physiographic and hydroclimatic	None	Relief (SRTM DEM 30m), Land Cover (MODIS

Figure	message	data used from simulation (variables) if any	data used from observation (if any)
	characteristics of the Sebou basin.		IGBP), Basin & River Network, In Situ Stations
Figure 2	Clearly describe the experimental protocol (Forcings -> LIS-Noah-MP -> OL vs DA -> HyMAP).	None	None
Figure 3	Identify corrections brought by DA according to seasons	Soil moisture layer 1 (0-5cm) from OL and DA	None

Figure	message	data used from simulation (variables) if any	data used from observation (if any)
	(winter vs summer).		
Figure 4	Demonstrate if assimilation indirectly captures the effect of irrigation and modifies the runoff.	Total Runoff (Q_s + Q_{sb}) from OL and DA	Irrigation Map (FAO GMIA v5)
Figure 5	Determine which hydrological component (surface runoff vs baseflow) is most sensitive to assimilation.	Surface runoff (Q_s), Baseflow (Q_{sb}), Soil Moisture from OL and DA	IMERG Precipitation

Figure	message	data used from simulation (variables) if any	data used from observation (if any)
Figures 6-9	Position the local LIS simulation results relative to global reference models.	Soil Moisture, Runoff, Evapotranspiration	GLDAS Products (NOAH, VIC, CLSM)
Figures 10-12	Quantify the improvement brought by DA on streamflow dynamics (hydrographs, KGE, NSE).	Routed discharge from OL and DA (HyMAP)	Gauging station discharge (In Situ)
Figures 13-14	Identify factors controlling the	Relative streamflow/runoff	IMERG Precipitation,

